The SU(2) bridge midpoint in closed form: the curvature term is exact

Proposition 6 of the series/parallel note described the midpoint of the Brownian bridge on $SU(2)$, the building block of the non-abelian parallel insertion, only to leading order in the heat time, with the curvature factor $1/h$, $h = (d/4)\cot(d/4)$, derived by Laplace's method and labelled as asymptotics. For the fundamental character the midpoint expectation is computable exactly. For a bridge of duration t from e to a group element at geodesic distance $\theta \in (0, 2\pi)$,

$$E\,\chi_{1/2}(m) = e^{-t/32}\Big[2\cos\frac{\theta}{4} - \frac{t}{2\theta}\,\sin\frac{\theta}{4}\Big] \text{ up to image terms of relative size } \Big(2+\frac{32\pi^2}{t}\Big)e^{-4\pi(2\pi-\theta)/t}$$

(Theorem 1). Expanded to first order it is exactly the prediction of Proposition 6(b), $2\cos\frac{\theta}{4}\,[1 - \frac{t}{32}(1 + \frac{2}{h(\theta)})]$, because $\frac{t}{32}\cdot\frac{2}{h} = \frac{t}{4\theta}\tan\frac{\theta}{4}$. So the curvature term N2, the side-flux-dependent softening of the mid-face weight, is confirmed with no remainder of order t^2 in the $w = 0$ sector: the only correction to the displayed formula consists of exponentially small nonzero Cartan-image terms. This gives an exact nonabelian statement about the parallel insertion in three dimensions, and it settles the fundamental-representation part of the uniform Laplace remainder that Proposition 7 of the series/parallel note assumes, for the trace in the fundamental representation (the image terms are of the stated order, and not uniformly so as $\theta \to 2\pi$).

The ingredients are character orthogonality and Poisson summation; the computation is elementary and no novelty is claimed for it.

1. Setting

Group metric as in the series/parallel note §1: $C_2(j) = j(j+1)$, $SU(2)$ the three-sphere of radius 2, heat kernel $k_t = \sum_j (2j+1)e^{-tj(j+1)/2}\chi_j$ with $\chi_j(\theta) = \sin((2j+1)\theta/2)/\sin(\theta/2)$ at rotation angle θ (the geodesic distance from e). The midpoint m of the bridge of duration t from e to g has density $k_{t/2}(m)\,k_{t/2}(m^{-1}g)/k_t(g)$ with respect to Haar measure. Its geodesic midpoint $m_* = \exp(\frac{1}{2}\log g)$ has rotation angle $\theta/2$, so $\chi_{1/2}(m_*) = \sin(\theta/2)/\sin(\theta/4) = 2\cos(\theta/4)$.

2. The exact formula

Theorem 1. Put $\Theta_\pm(\theta) = \sum_{w\in\mathbb{Z}}(\pm1)^w e^{-(\theta-4\pi w)^2/(2t)}$ and $\Xi_\pm(\theta) = \sum_{w\in\mathbb{Z}}(\pm1)^w(\theta - 4\pi w)e^{-(\theta-4\pi w)^2/(2t)}$. For $\theta \in (0, 2\pi)$,

$$E\,\chi_{1/2}(m) = e^{-t/32}\,\frac{2\cos(\theta/4)\,\Xi_-(\theta) - \frac{t}{2}\sin(\theta/4)\,\Theta_-(\theta)}{\Xi_+(\theta)}. \tag{1}$$

Keeping only $w = 0$ gives the displayed formula of the summary.

Proof. Expand both heat kernels in characters. Character orthogonality in the form $\int \chi_a(m)\chi_b(m^{-1}g)\,dm = \delta_{ab}\chi_a(g)/d_a$, together with $\chi_{j_1}\chi_{1/2} = \chi_{j_1+1/2} + \chi_{j_1-1/2}$, gives

$$k_t(g)\,E\,\chi_{1/2}(m) = \sum_{j_1}\ \sum_{j_2=j_1\pm1/2} (2j_1+1)\,e^{-t[C_2(j_1)+C_2(j_2)]/4}\,\chi_{j_2}(g).$$

Write $n = 2j_1 + 1$. Then $C_2(j_1) + C_2(j_2) = \frac{1}{4}[n^2 + (n\pm1)^2 - 2] = \frac{1}{2}(n\pm\frac{1}{2})^2 - \frac{3}{8}$, so the exponential is $e^{3t/32}e^{-t(n\pm1/2)^2/8}$. Multiply by $\sin(\theta/2)$, reindex the $j_2 = j_1 - \frac{1}{2}$ sum by $n \mapsto n+1$, and write $s = n + \frac{1}{2}$ over $s \in \frac{1}{2} + \mathbb{N}$: the summand becomes $(s-\frac{1}{2})\sin\frac{(s+\frac{1}{2})\theta}{2} + (s+\frac{1}{2})\sin\frac{(s-\frac{1}{2})\theta}{2} = 2s\cos\frac{\theta}{4}\sin\frac{s\theta}{2} - \sin\frac{\theta}{4}\cos\frac{s\theta}{2}$, which is even in s. Hence

$$k_t(g)\sin\frac{\theta}{2}\;E\,\chi_{1/2}(m) = \frac{e^{3t/32}}{2}\Big[2\cos\frac{\theta}{4}\,A - \sin\frac{\theta}{4}\,B\Big],$$

with $B = \sum_{s\in\frac12+\mathbb{Z}} e^{-ts^2/8}\cos\frac{s\theta}{2}$ and $A = \sum_{s\in\frac12+\mathbb{Z}} s\,e^{-ts^2/8}\sin\frac{s\theta}{2} = -2\partial_\theta B$. In the same way $k_t(g)\sin\frac{\theta}{2} = \frac{e^{t/8}}{2}A_0$ with $A_0 = \sum_{n\in\mathbb{Z}} n\,e^{-tn^2/8}\sin\frac{n\theta}{2}$. Poisson summation over the half-integers (which introduces the sign $(-1)^w$) and over the integers gives $B = \sqrt{8\pi/t}\,\Theta_-$, $A = (2/t)\sqrt{8\pi/t}\,\Xi_-$ and $A_0 = (2/t)\sqrt{8\pi/t}\,\Xi_+$. Divide. □

The image terms. For $\theta \in (0, 2\pi)$ the pair $w = \pm1$ in $\Xi_\pm$ has, relative to the $w = 0$ term $\theta e^{-\theta^2/(2t)}$, the size $e^{-8\pi^2/t}\,|2\theta\cosh(4\pi\theta/t) - 8\pi\sinh(4\pi\theta/t)|/\theta \le (2 + 32\pi^2/t)\,e^{-4\pi(2\pi-\theta)/t}$, using $\sinh x \le x\cosh x$. In Θ_- the same pair has relative size at most $2e^{-4\pi(2\pi-\theta)/t}$. The pairs $|w| \ge 2$ are smaller by a further factor $e^{-16\pi^2/t}$ up to polynomial factors. This is the large-field term N4, the winding of the bridge around the group, and it is negligible uniformly on $\theta \le \theta_0 < 2\pi$.

3. Comparison with Proposition 6

Proposition 6(b) predicts that $m = m_* e^\xi$ with ξ Gaussian of mean zero and covariance $\frac{t}{4}\mathrm{diag}(1, \frac{1}{h}, \frac{1}{h})$ in the frame (axis, transverse). In the spin-$\frac12$ representation $(\xi\cdot T)^2 = -\frac14|\xi|^2$, so $E\,D^{1/2}(e^\xi) = 1 - \frac18 E|\xi|^2 + O(t^2) = 1 - \frac{t}{32}(1 + \frac{2}{h}) + O(t^2)$, and $E\,\chi_{1/2}(m) = 2\cos\frac{\theta}{4}\,[1 - \frac{t}{32}(1+\frac{2}{h})] + O(t^2)$. The $w = 0$ part of (1) equals $2\cos\frac{\theta}{4}\,e^{-t/32}[1 - \frac{t}{4\theta}\tan\frac{\theta}{4}]$, and $\frac{t}{32}\cdot\frac{2}{h(\theta)} = \frac{t}{16}\cdot\frac{\tan(\theta/4)}{\theta/4} = \frac{t}{4\theta}\tan\frac{\theta}{4}$. The two agree at first order, curvature factor included. As $\theta \to 0$ both give $2(1 - \frac{3t}{32})$, the flat value $E|\xi|^2 = \frac{3t}{4}$.

Formula (1) says more than the asymptotics. In the $w = 0$ sector the fundamental character of the midpoint is the product of an isotropic factor $e^{-t/32}$ and the exactly linear curvature correction $1 - \frac{t}{4\theta}\tan\frac{\theta}{4}$; there are no higher-order terms in t. The correction grows as $\theta \to 2\pi$, where $\tan(\theta/4)$ diverges and the midpoint becomes bimodal, and there the image terms take over. This is the non-abelian counterpart of the parity factor $\rho_t(\phi)$ of the $U(1)$ cube (Proposition 3 of the series/parallel note).

4. Every spin

Theorem 2. For every spin J and $\theta \in (0, 2\pi)$, with $\sigma = 0$ for integer J and $\sigma = \frac12$ for half-integer J,

$$E\,\chi_J(m) = \frac{\Xi_{(\sigma)}(\theta)}{\Xi_+(\theta)}\sum_{k=-J}^{J} e^{-tk^2/8}\cos\frac{k\theta}{2} - \frac{t}{2}\,\frac{\Theta_{(\sigma)}(\theta)}{\Xi_+(\theta)}\sum_{k=-J}^{J} k\,e^{-tk^2/8}\sin\frac{k\theta}{2},$$

where $\Xi_{(0)} = \Xi_+$, $\Theta_{(0)} = \Theta_+$, $\Xi_{(1/2)} = \Xi_-$, $\Theta_{(1/2)} = \Theta_-$, and k runs in unit steps. In the $w = 0$ sector,

$$E\,\chi_J(m) = \sum_{k=-J}^{J} e^{-tk^2/8}\Big[\cos\frac{k\theta}{2} - \frac{t}{2\theta}\,k\sin\frac{k\theta}{2}\Big]. \tag{2}$$

For integer J the first ratio is exactly one, image terms included.

Proof. As for Theorem 1, now with $\chi_{j_1}\chi_J = \sum_{k=-J}^{J}\chi_{j_1+k}$. This identity holds for every j_1 if χ_j is defined by $\sin((2j+1)\theta/2)/\sin(\theta/2)$ also for $2j+1 \le 0$: it is the product-to-sum formula for $\sin(n\theta/2)\sum_k e^{ik\theta}$. The formal terms with $2j_2 + 1 \le 0$ cancel in pairs carrying the same weight, so the sum may be taken over all $n = 2j_1 + 1 \in \mathbb{Z}$, halved; the summand is even under $(n, k) \mapsto (-n, -k)$. With $n_2 = n + 2k$, $C_2(j_1) + C_2(j_2) = \frac12(n+k)^2 + \frac12(k^2 - 1)$, so the weight is $e^{t/8}e^{-tk^2/8}e^{-ts^2/8}$ with $s = n + k \in \mathbb{Z} + \sigma$. Expanding $(s-k)\sin\frac{(s+k)\theta}{2}$ and discarding the terms odd in s leaves $\cos\frac{k\theta}{2}\cdot s\sin\frac{s\theta}{2} - k\sin\frac{k\theta}{2}\cos\frac{s\theta}{2}$. Summing over s gives A_σ and B_σ, Poisson summation gives $\Xi_{(\sigma)}$ and $\Theta_{(\sigma)}$, and the factor $e^{t/8}$ cancels against the denominator. For $J = \frac12$ this is Theorem 1. □

Agreement with Proposition 6 for every spin. The leading term of (2) is $\sum_k \cos(k\theta/2) = \chi_J(m_*)$, the character at the geodesic midpoint (rotation angle $\theta/2$). The

Gaussian prediction of Proposition 6(b) at first order is $\sum_k \cos\frac{k\theta}{2}\,[1-\frac{t}{8}k^2-\frac{t}{8}(J(J+1)-k^2)/h]$ with $1/h = 4\tan(\theta/4)/\theta$. The two agree iff, with $\varphi=\theta/2$,

$$\sum_k k\sin(k\varphi) = \tan\frac{\varphi}{2}\sum_k \big(J(J+1)-k^2\big)\cos(k\varphi).$$

The left side is $-\chi_J'(\varphi)$ and the sum on the right is $J(J+1)\chi_J+\chi_J''$. The class-function Laplacian of $SU(2)$ in this metric gives $\chi_J''+\cot(\varphi/2)\,\chi_J' = -J(J+1)\chi_J$, which is the identity. (For $J=\frac12$ and $J=1$ it can be checked directly: $\sin(\varphi/2)=\tan(\varphi/2)\cos(\varphi/2)$ and $2\sin\varphi = \tan(\varphi/2)(2+2\cos\varphi)$.) So the curvature term of Proposition 6 is exact at first order in every representation, and (2) contains no terms of order t^2 beyond the Gaussian factors $e^{-tk^2/8}$.

What the trace does not give. By Proposition 6(a), $E\,D^J(m) = D^J(m_*)\Lambda^J$ with Λ^J real diagonal in the axis basis, and (2) fixes only $\sum_\mu e^{i\mu\theta/2}\Lambda^J_{\mu\mu}$, one equation for $2J+1$ entries. The isolated cube needs the entries, because its four cut faces have different axes.

5. The fundamental sector of the isolated cube is exact

Theorem 3. (a) In spin $\frac12$ the midpoint's matrix expectation is a real multiple of a unitary: $E\,D^{1/2}(m) = \lambda(\theta,t)\,D^{1/2}(m_*)$ with

$$\lambda(\theta,t) = e^{-t/32}\,\frac{\Xi_-(\theta)-\frac{t}{4}\tan\frac{\theta}{4}\,\Theta_-(\theta)}{\Xi_+(\theta)} \;\overset{w\equiv 0}{=}\; e^{-t/32}\Big[1-\frac{t}{4\theta}\tan\frac{\theta}{4}\Big], \qquad \theta\in(0,2\pi).$$

(b) For one refined cube (series/parallel note, §4), whose four cut side faces have heat times t_j and side angles θ_j, the spin-$\frac12$ term of the character sum for Ψ is exactly

$$d_{1/2}\,e^{-3t_m/8}\,\mathrm{tr}_{1/2}\prod_j E\,D^{1/2}(m_j^{\pm1}) = 2\,e^{-3t_m/8}\Big(\prod_{j=1}^4\lambda(\theta_j,t_j)\Big)\chi_{1/2}(h_{\mathrm{int}}),$$

where h_{int} is the holonomy of the geodesic interpolation around the mid-face. In this sector the interpolation term N1 is exact (it is h_{int}), the curvature term N2 is exact (it is $\prod\lambda$), and the commutator term N3 vanishes.

Proof. (a) By Proposition 6(a) of the series/parallel note, $E\,D^{1/2}(m) = D^{1/2}(m_*)\Lambda$ with $\Lambda=\mathrm{diag}(\Lambda_+,\Lambda_-)$ real in the axis basis, where $D^{1/2}(m_*)=\mathrm{diag}(e^{i\theta/4},e^{-i\theta/4})$. Its trace $e^{i\theta/4}\Lambda_+ + e^{-i\theta/4}\Lambda_-$ equals $E\,\chi_{1/2}(m)$, which is real because the midpoint density and $\chi_{1/2}$ are real. The imaginary part gives $\sin(\theta/4)(\Lambda_+-\Lambda_-)=0$, so $\Lambda_+=\Lambda_-=\lambda$, and $\lambda = E\,\chi_{1/2}(m)/(2\cos(\theta/4))$ by Theorem 1. (b) The bridge for a cut face runs from Q_j^{-1} to P_j; left translation by Q_j^{-1} maps it to a bridge from e, so $E\,D^{1/2}(m_j) = \lambda_j D^{1/2}(m_{*j})$ with m_{*j} the geodesic midpoint, and $E\,D^{1/2}(m_j^{-1}) = \lambda_j D^{1/2}(m_{*j})^{-1}$ because λ_j is real. The four midpoints are independent, since each is fixed by its own cut face and an isolated cube has no shared mid-edges, so the trace of the product of expectations is $\prod_j\lambda_j$ times the trace of the product of the unitaries, which is $\chi_{1/2}(h_{\mathrm{int}})$. The prefactor is $d_{1/2}e^{-t_m C_2(1/2)/2}$ with $C_2(\frac12)=\frac34$. □

For $J\ge 1$ the diagonal entries differ: the Gaussian prediction is $\Lambda^J_{\mu\mu}\simeq 1-\frac{t}{8}[\mu^2+(J(J+1)-\mu^2)/h]$, which depends on μ unless $h=1$. The Weyl reflection combined with time reversal gives $\Lambda^J_{\mu\mu}=\Lambda^J_{-\mu,-\mu}$ for every J: with w in the normaliser of the torus, $\mathrm{Ad}(w)X=-X$, the map $g\mapsto e^X w g w^{-1}$ carries the bridge from e to e^X to the bridge from e^X to e, whose midpoint law is the same by reversibility, and conjugation by $D(w)$ exchanges μ and $-\mu$ on a diagonal matrix (reviewer's argument; the reality of the trace alone gives only one relation). The trace (2) then fixes $\lfloor J\rfloor+1$ unknowns by one equation, so for $J=1$ one relation remains between Λ^1_{11} and Λ^1_{00}. The matrix-element version of the orthogonality computation of Theorem 1, with the Clebsch–Gordan coefficients of $j_1\otimes J$, is the tool for them.

6. The spin-1 matrix and isolated cube

Theorem 4 (spin-1 midpoint matrix; internal written checks). Notation of this section: E in the image bounds, A_0, A_1, $B(x)$ and P_j are local to Theorem 4 and differ from the quantities of the same names in Theorems 1 and 3. The spin-1 midpoint matrix is exactly determined by two real entries $\lambda_0 = \Lambda^1_{00}$ and $\lambda_1 = \Lambda^1_{11} = \Lambda^1_{-1,-1}$. The formulas below give both entries, including winding, for every $t > 0$ and $0 < \theta < 2\pi$. They involve a Gaussian tail, which retains higher-order curvature corrections beyond the character formula.

Put $x = \theta/2$, $\eta = e^{t/8}$, and define

$$G_t(z) = e^{-2z^2/t}, \qquad T_t(z) = \int_z^\infty (1 - \eta\cos u)e^{-2u^2/t}\,du, \qquad S_t(x) = \sum_{w\in\mathbb{Z}} T_t(x - 2\pi w). \tag{3}$$

The Gaussian integral of $1 - \eta\cos u$ over the real line is zero; thus T_t is odd and decays at both ends. In particular the image sum in (3) is absolutely convergent. A special-function closed form is

$$T_t(z) = \sqrt{\frac{\pi t}{8}}\left[\operatorname{erfc}\left(\sqrt{\frac{2}{t}}\,z\right) - \operatorname{Re}\operatorname{erfc}\left(\sqrt{\frac{2}{t}}\,z - i\sqrt{\frac{t}{8}}\right)\right]. \tag{4}$$

With $\Theta = \Theta_+(\theta)$ and $\Xi = \Xi_+(\theta)$ of Theorem 1,

$$\boxed{\lambda_0 = 1 + \frac{t\eta^{-1}}{\sin x\,\Xi}\left[(\eta\cos x - 1)\Theta - \cot x\, S_t(x)\right],} \tag{5}$$

$$\boxed{\lambda_1 = \eta^{-1}\left\{1 - \frac{t}{2\sin x\,\Xi}\left[(\eta - \cos x)\Theta - \frac{S_t(x)}{\sin x}\right]\right\}.} \tag{6}$$

These expressions are regular at $\theta = \pi$. The endpoint $\theta = 0$ is obtained by continuity and has $\lambda_0 = \lambda_1$ by conjugation invariance. The denominator Ξ is positive on the stated open interval, since it is proportional to $k_t(g)\sin x$.

Matrix-element computation. Write $c_{a,b}(\alpha,\mu) = |\langle a\alpha; 1\mu \mid b, \alpha+\mu\rangle|^2$. The spin-1 Clebsch–Gordan squares are

b	$c_{a,b}(\alpha,0)$	$c_{a,b}(\alpha,1)$
$a+1$	$\dfrac{(a+1)^2-\alpha^2}{(a+1)(2a+1)}$	$\dfrac{(a+\alpha+1)(a+\alpha+2)}{2(a+1)(2a+1)}$
a	$\dfrac{\alpha^2}{a(a+1)}$	$\dfrac{(a-\alpha)(a+\alpha+1)}{2a(a+1)}$
$a-1$	$\dfrac{a^2-\alpha^2}{a(2a+1)}$	$\dfrac{(a-\alpha)(a-\alpha-1)}{2a(2a+1)}$

Here $c_{a,b}(\alpha,-1) = c_{a,b}(-\alpha,1)$, and only actual summands of $a\otimes 1$ are included: $a = 0$ has just $b = 1$, and $a = 1/2$ has $b = 1/2, 3/2$. These squares follow by lowering the highest vector of spin $a+1$ and taking the orthogonal spin-a and spin-$(a-1)$ vectors at each total weight. The lowering rule used is $J_-|a,\alpha\rangle = \sqrt{(a+\alpha)(a-\alpha+1)}|a,\alpha-1\rangle$; normalization gives the denominators in the table.

Matrix-element orthogonality gives, with $C_a = a(a+1)$,

$$k_t(g)E[D^1(m)]_{\mu\mu} = \sum_{a,b}(2a+1)e^{-t(C_a+C_b)/4}\sum_{\alpha=-a}^{a} c_{a,b}(\alpha,\mu)e^{i(\alpha+\mu)\theta}. \tag{7}$$

The factor $2b+1$ in the second heat kernel cancels its orthogonality denominator. For $\mu = 0$, put $N_0 = k_t(g)\lambda_0$ and

$$B(x) = \sum_{n\in\mathbb{Z}} e^{-tn^2/8}\cos(nx), \qquad H(x) = \sum_{n\neq 0} e^{-tn^2/8}\frac{\sin(nx)}{n}, \qquad H(x) = \int_0^x (B(y)-1)\,dy.$$

We compute the $b = a \pm 1$ and $b = a$ parts of (7) separately. In the first two parts multiplication by α^2 becomes $-\partial_\theta^2$. Write $n = 2a+1$ and shift to $s = n \pm 1$; the common heat weight is $e^{-ts^2/8}$. Combining the two parts gives

$$\begin{aligned} N_{\text{off}} &= \sum_{s\geq 1} e^{-ts^2/8}\left(s + \frac{1}{s}\partial_x^2\right)[\cot x \sin(sx)] \\ &= \frac{\cot x\, H(x) - B(x) + 1}{\sin^2 x}. \end{aligned} \tag{8}$$

The formal lower-spin term at $a = 1/2$ vanishes because $(a^2 + \partial_\theta^2)\chi_a = 0$; the missing upper partner at $s = 1$ has $\sin((s-1)x) = 0$. This accounts for the endpoints of the reindexing. The cancellation in (8) follows directly from

$$\partial_x^2[\cot x \sin(sx)] = 2\csc^2 x \cot x \sin(sx) - 2s\csc^2 x \cos(sx) - s^2 \cot x \sin(sx).$$

For the diagonal coupling $b = a$, set $f_n(x) = \sin(nx)/\sin x$. Its elementary differential identity is $f_n'' + 2\cot x\, f_n' = -(n^2-1)f_n$. Hence

$$N_{\text{diag}} = k_t(g) - 1 + 2\eta\cot x\left(\frac{V(x)}{\sin x}\right)', \qquad V(x) = \sum_{n\geq 2}\frac{ne^{-tn^2/8}}{n^2-1}\sin(nx).$$

The omitted $n = 1$ term in the heat kernel is exactly 1. Moreover, $V'' + V = B'/2 + \eta^{-1}\sin x$ and $V(0) = 0$, so the Wronskian identity gives

$$\left(\frac{V}{\sin x}\right)' = \frac{1}{\sin^2 x}\int_0^x \sin y\left(\frac{B'(y)}{2} + \eta^{-1}\sin y\right)dy.$$

Adding (8), integrating $\sin y B'(y)$ once, and using $2\int_0^x \sin^2 y\,dy = x - \sin x\cos x$ yields

$$N_0 = k_t(g) + \frac{1}{\sin^2 x}\left[(\eta\cos x - 1)B(x) + \cot x\int_0^x (1-\eta\cos y)B(y)\,dy\right]. \tag{9}$$

Every series above and its displayed derivatives converge absolutely for $t > 0$. Poisson summation, with the normalization of Theorem 1, gives

$$B(x) = \sqrt{\frac{8\pi}{t}}\sum_w G_t(x - 2\pi w), \qquad k_t(g) = \frac{\eta}{t\sin x}\sqrt{\frac{8\pi}{t}}\,\Xi.$$

Periodicity of the cosine and oddness of T_t give

$$\int_0^x (1-\eta\cos y)B(y)\,dy = -\sqrt{\frac{8\pi}{t}}\,S_t(x) :$$

the integration constants are $\sum_w T_t(-2\pi w) = 0$, paired in w. Substitution into (9) proves (5).

Weyl reflection with time reversal, as proved in §5, gives $\Lambda^1_{11} = \Lambda^1_{-1,-1}$. Theorem 2 therefore reads

$$\lambda_0 + 2\cos x\,\lambda_1 = 1 + 2\eta^{-1}\cos x - t\eta^{-1}\sin x\frac{\Theta}{\Xi}. \tag{10}$$

Inserting (5), the numerator for $2\cos x\,\lambda_1$ factors by $\cos x$, giving (6). This first proves (6) when $\cos x \neq 0$; continuity proves it also at $x = \pi/2$. Here (6) is obtained from (5) through (10), and (10) itself follows from (7) because $\sum_\mu c_{a,b}(m-\mu,\mu) = 1$; so the trace is a consistency of the framework. Reality and the equality of the opposite weights are explicit. □

Independent check through the $\mu = 1$ column (Fable referee, 2026-09-27). Reducing (7) directly at $\mu = 1$, with total weight m, $p = 2m$ and $q = p-1$: the $b = a \pm 1$ parts combine to the summand $(s+p)(s+p-2)+(s-p)(s-p+2) = 2s^2+2q^2-2$, where q runs over the weights of spin $(s-1)/2$, so the off-diagonal part is $\sum_{s\geq 1}(e^{-ts^2/8}/s)[(s^2-1)f_s + \cot x\, f_s']$ with f_s the character sum of spin $(s-1)/2$; the diagonal part has bracket $2f_n'(i - \cot x) = -2f_n' e^{-ix}/\sin x$. Assembling with $H' = B - 1$ and the Wronskian of the diagonal step reproduces (6) exactly. Since (6) was derived from (5) and (10), this confirms (5) as well. (Before (9), the -1 of the diagonal part is absorbed by $1 - \cos^2 x = \sin^2 x$.) At first order $\lambda_0 - \lambda_1 = -\frac{t}{8}(\frac{1}{h} - 1) < 0$, since $h < 1$ on $0 < \theta < 2\pi$.

The zero image and explicit image bounds. Define $R_t(x) = e^{2x^2/t}T_t(x)$ and

$$A_0 = \eta\cos x - 1 - \cot x\, R_t(x), \qquad A_1 = \eta - \cos x - \frac{R_t(x)}{\sin x}.$$

Keeping $w = 0$ in both numerator and denominator of (5)–(6) gives

$$\lambda_0^{(0)} = 1 + \frac{t\eta^{-1}}{\theta\sin x}A_0, \qquad \lambda_1^{(0)} = \eta^{-1}\left[1 - \frac{t}{2\theta\sin x}A_1\right]. \tag{11}$$

For explicit bounds on all remaining images, put

$$q = e^{-4\pi(2\pi-\theta)/t}, \qquad r = e^{-16\pi^2/t}, \qquad E = \frac{2q}{1-r},$$

$$\delta = q\left[\frac{2}{1-r} + \frac{32\pi^2}{t}\frac{1+r}{(1-r)^3}\right], \qquad M = \frac{(1+\eta)t}{4(2\pi - x)}.$$

Whenever $\delta < 1$,

$$|\lambda_0 - \lambda_0^{(0)}| \leq \frac{t\eta^{-1}}{\theta\sin x(1-\delta)}\left\{E\left[|\eta\cos x - 1| + |\cot x|M\right] + \delta|A_0|\right\}, \tag{12}$$

$$|\lambda_1 - \lambda_1^{(0)}| \leq \frac{t\eta^{-1}}{2\theta\sin x(1-\delta)}\left\{E\left[\eta - \cos x + \frac{M}{\sin x}\right] + \delta|A_1|\right\}. \tag{13}$$

To verify these bounds, write $G = G_t(x)$. The pair $w = \pm n$ has Gaussian ratio bounded by $2q_n$, where $q_n = e^{-8\pi^2n^2/t + 8\pi nx/t} \leq qr^{n-1}$. Its contribution to $\Xi/(\theta G) - 1$ is at most $(2 + 32\pi^2n^2/t)q_n$, by the same $\sinh u \leq u\cosh u$ estimate as in Theorem 1. Consequently

$$\left|\frac{\Theta}{G} - 1\right| \leq E, \qquad \left|\frac{\Xi}{\theta G} - 1\right| \leq \delta.$$

For $z \neq 0$, oddness and the Gaussian tail bound give

$$|T_t(z)| \leq (1+\eta)\int_{|z|}^\infty G_t(u)\,du \leq \frac{(1+\eta)t}{4|z|}G_t(z).$$

All nonzero images have $|x - 2\pi w| \geq 2\pi - x$, whence $|S_t(x)/G - R_t(x)| \leq ME$. Dividing by $\Xi/(\theta G)$ proves (12)–(13), including the denominator error. The formulas (5)–(6) hold at all $t > 0$; the explicit small-image estimates require the displayed condition $\delta < 1$. They are exponentially small uniformly on each compact subinterval of $0 < \theta < 2\pi$ as $t \downarrow 0$. Their displayed prefactors become loose at $\theta = 0$, where the exact expressions have removable singularities; their exponential separation degenerates at the cut locus $\theta = 2\pi$.

Gaussian and central-endpoint checks. Integration by parts in (3), at fixed $x > 0$, gives

$$R_t(x) = \frac{t}{4x}(1 - \eta\cos x) + O_x(t^2).$$

Thus, with $h = (x/2)\cot(x/2)$,

$$\lambda_0 = 1 - \frac{t}{4h} + O_\theta(t^2), \qquad \lambda_1 = 1 - \frac{t}{8}\left(1 + \frac{1}{h}\right) + O_\theta(t^2). \tag{14}$$

Here the exponentially small image errors are absorbed into the remainder. This is exactly $1 - \frac{t}{8}[\mu^2 + (2-\mu^2)/h]$ for $\mu = 0, \pm 1$. At $\theta = 0$, conjugation invariance and (10) give the independent check

$$\lambda_0(0,t) = \lambda_1(0,t) = \frac{1}{3}\left[1 + 2\eta^{-1} - \frac{t\eta^{-1}}{2}\frac{\Theta_+(0)}{\partial_\theta \Xi_+(0)}\right].$$

The zero-image limit of either expression (11) is $[1 + 2\eta^{-1} - (t/2)\eta^{-1}]/3 = 1 - t/4 + O(t^2)$, in agreement with $h(0) = 1$.

Consequence for the isolated cube. For cut face j, denote the bridge endpoints by A_j, B_j, put $X_j = \log(A_j^{-1}B_j)$, and set $m_{*j} = A_j e^{X_j/2}$. Let P_j project onto weight zero along X_j in spin 1. In the three-dimensional vector realization it is the rank-one projector $\hat{X}_j\hat{X}_j^{\mathsf{T}}$. With (5)–(6) at (θ_j, t_j), put

$$L_j = \lambda_{1j} I + (\lambda_{0j} - \lambda_{1j})P_j, \qquad U_j = D^1(m_{*j}), \qquad F_j^+ = U_j L_j, \quad F_j^- = L_j U_j^{-1}.$$

The four independent bridges of one isolated cube give its exact spin-1 contribution to Ψ:

$$\boxed{\Psi_1 = 3e^{-t_m}\,\mathrm{tr}_1\left(F_1^{\varepsilon_1}F_2^{\varepsilon_2}F_3^{\varepsilon_3}F_4^{\varepsilon_4}\right), \qquad \varepsilon_j \in \{+,-\},} \tag{15}$$

in the boundary order and orientations of the mid-face. The exponent is $-t_m C_2(1)/2 = -t_m$. Left translation reduces each bridge to (5)–(6); inversion takes the adjoint and explains the order in F_j^-. The independence is the same isolated-cube hypothesis as in Theorem 3.

This sector retains an exact anisotropy $\lambda_0 - \lambda_1 = (t/8)(1 - 1/h) + O_\theta(t^2)$. After transporting axes to a common frame, two such factors obey

$$[L_i, L_j] = (\lambda_{0i} - \lambda_{1i})(\lambda_{0j} - \lambda_{1j})[P_i, P_j], \qquad \|[P_i, P_j]\| = |c|\sqrt{1-c^2} \le \tfrac{1}{2},$$

where c is the scalar product of the transported unit axes. Consequently the axis-dependent matrix product in (15) supplies the spin-1 noncommuting correction explicitly; generic oblique axes give a commutator of order $t_i t_j$ at fixed nonzero side angles. The scalar factorization of Theorem 3 is special to spin 1/2.

For an explicit winding bound on (15), let ϵ_j be the maximum of the right sides of (12)–(13) for face j, assuming each $\delta_j < 1$, and replace each L_j by its zero-image value to obtain $\Psi_1^{(0)}$. Since an expectation of unitary matrices has norm at most 1, telescoping the four products gives

$$|\Psi_1 - \Psi_1^{(0)}| \le 9e^{-t_m}\left[\prod_{j=1}^{4}(1+\epsilon_j) - 1\right]. \tag{16}$$

The factor 9 is the character coefficient 3 times the trace-norm bound $|\,\mathrm{tr}_1 A| \le 3\|A\|$.

7. Consequence for STATE

Atlas cell 1 is filled through spin 1 by Theorem 4, with exact matrix entries, explicit image bounds and the isolated-cube contribution. For $J > 1$, (7) with the spin-J Clebsch–Gordan squares is still an exact spectral representation. The finite Gaussian-tail reduction and its bounds have been completed here for $J = 1$; the higher-spin entries and estimates permitting the full character sum remain the next obligation before the isolated-cube Laplace claim of Proposition 7. The full mid-plane and iteration remain the subsequent atlas cells.